

1 Curvature of a Curve

Idea: *Curvature* measures the extent to which a curve is not contained in a straight line.

1.1 Curvature

Motivation: Let $\gamma : I \rightarrow \mathbb{R}^2$ be a plane curve with unit-speed, i.e., $\|\dot{\gamma}(t)\| \equiv 1$, we wish to find a way to measure how *curved* this plane curve is at some point $\mathbf{p}$, this should be the deviation from the tangent line at $\mathbf{p}$. Let $\gamma(t_0)$ be a point on the curve, when the parameter t changes from t_0 to $t_0 + \Delta t$, the distance that γ moves away from the tangent line at $\gamma(t_0)$ can be calculated as $(\gamma(t_0 + \Delta t) - \gamma(t_0)) \cdot \mathbf{n}(t_0)$, where $\mathbf{n}(t_0)$ is the unit-vector such that $\mathbf{n}(t_0) \perp \dot{\gamma}(t_0)$ at the point $\gamma(t_0)$.

By Taylor's theorem, we have

$$\gamma(t_0 + \Delta t) = \gamma(t_0) + \dot{\gamma}(t_0)\Delta t + \frac{1}{2}\ddot{\gamma}(t_0)(\Delta t)^2 + o((\Delta t)^2) \quad (\Delta t \rightarrow 0),$$

since $\dot{\gamma}(t_0) \cdot \mathbf{n}(t_0) = 0$, hence we get

$$\begin{aligned} (\gamma(t_0 + \Delta t) - \gamma(t_0)) \cdot \mathbf{n}(t_0) &= \left(\dot{\gamma}(t_0)\Delta t + \frac{1}{2}\ddot{\gamma}(t_0)(\Delta t)^2 + o((\Delta t)^2) \right) \cdot \mathbf{n}(t_0) \\ &= \frac{1}{2}\ddot{\gamma}(t_0)(\Delta t)^2 \cdot \mathbf{n}(t_0) + o((\Delta t)^2) \cdot \mathbf{n}(t_0) \quad (\Delta t \rightarrow 0) \end{aligned}$$

then the deviation of γ from its tangent line at $\gamma(t_0)$ is given by

$$\left\| (\gamma(t_0 + \Delta t) - \gamma(t_0)) \cdot \mathbf{n}(t_0) \right\| = \left\| \frac{1}{2}\ddot{\gamma}(t_0)(\Delta t)^2 \cdot \mathbf{n}(t_0) + o((\Delta t)^2) \cdot \mathbf{n}(t_0) \right\|,$$

so if we neglect the remainder terms, the deviation is $\frac{1}{2}\|\ddot{\gamma}(t_0)\|(\Delta t)^2$. So this implies that $\|\ddot{\gamma}(t_0)\|$ can be used to measure the curvature of γ .

Definition 1.(Curvature of a Curve)

Let $\gamma(t)$ be a unit-speed curve in $\mathbb{R}^n$, i.e., $\|\dot{\gamma}(t)\| \equiv 1$. The *curvature* $\kappa(t)$ at the point $\gamma(t_0)$ is defined by

$$\kappa(t_0) := \|\ddot{\gamma}(t_0)\|.$$

Remark: Let $\hat{\gamma}$ and $\tilde{\gamma}$ be two different unit-speed reparametrization of γ , then by the property of unit-speed reparametrization, we know that

$$\tilde{\gamma}(t) = \hat{\gamma}(u) \text{ with } u(t) = \pm t + c \text{ where } c \text{ is a constant,}$$

thus we get

$$\begin{aligned} \frac{d\tilde{\gamma}}{dt} &= \frac{d\hat{\gamma}}{du} \cdot \frac{du}{dt} = \pm \frac{d\hat{\gamma}}{du}, \\ \frac{d^2\tilde{\gamma}}{dt^2} &= \frac{d}{dt} \left(\pm \frac{d\hat{\gamma}}{du} \right) = \pm \frac{d}{du} \left(\frac{d\hat{\gamma}}{du} \right) \cdot \frac{du}{dt} = \frac{d^2\hat{\gamma}}{du^2}, \end{aligned}$$

which means that $\tilde{\kappa} \equiv \hat{\kappa}$, hence curvature κ is well-defined.

Example 1.(Curvature of a Circle)

Consider the circle

$$\gamma(t) := \left(x_0 + R \cos \frac{t}{R}, y_0 + R \sin \frac{t}{R} \right),$$

then $\kappa(t) \equiv \frac{1}{R}$.

Proof: since

$$\dot{\gamma}(t) = \left(-\sin \frac{t}{R}, \cos \frac{t}{R} \right),$$

so we have $\|\dot{\gamma}(t)\| \equiv 1$,

$$\ddot{\gamma}(t) = \left(-\frac{1}{R} \cos \frac{t}{R}, -\frac{1}{R} \sin \frac{t}{R} \right),$$

so by the definition, we get

$$\kappa(t) = \|\ddot{\gamma}(t)\| = \frac{1}{R}.$$

Remark: This is accordance with the expectation that a small circle should have large curvature.

The following proposition gives us a formula for calculating the curvature of any regular curve in $\mathbb{R}^3$.

Proposition 1.(Curvature of a Regular Curve in $\mathbb{R}^3$)

Let $\gamma(t)$ be a regular curve in $\mathbb{R}^3$, then its curvature is given by

$$\kappa(t) = \frac{\|\ddot{\gamma}(t) \times \dot{\gamma}(t)\|}{\|\dot{\gamma}(t)\|^3}.$$

Proof: Let $s(t) = \int_{t_0}^t \|\dot{\gamma}(u)\| du$ be the arc-length parameter of γ , then we have

$$\dot{\gamma}(t) = \frac{d\gamma}{dt} = \frac{d\gamma}{ds} \cdot \frac{ds}{dt},$$

hence the curvature is

$$\kappa = \left\| \frac{d^2\gamma}{ds^2} \right\| = \left\| \frac{d}{ds} \left(\frac{d\gamma/dt}{ds/dt} \right) \right\|,$$

and note that

$$\frac{d}{ds} \left(\frac{d\gamma/dt}{ds/dt} \right) = \frac{\frac{d}{dt} \left(\frac{d\gamma/dt}{ds/dt} \right)}{ds/dt} = \frac{\frac{d^2\gamma}{dt^2} \cdot \frac{ds}{dt} - \frac{d\gamma}{dt} \cdot \frac{d^2s}{dt^2}}{(ds/dt)^3},$$

thus

$$\kappa = \left\| \frac{\frac{d^2\gamma}{dt^2} \cdot \frac{ds}{dt} - \frac{d\gamma}{dt} \cdot \frac{d^2s}{dt^2}}{(ds/dt)^3} \right\|.$$

Since $\frac{ds}{dt} = \|\dot{\gamma}(t)\|$, so

$$\left(\frac{ds}{dt} \right)^2 = \|\dot{\gamma}(t)\|^2 = \dot{\gamma}(t) \cdot \dot{\gamma}(t),$$

hence we have

$$\frac{d}{dt} \left(\left(\frac{ds}{dt} \right)^2 \right) = 2 \frac{ds}{dt} \frac{d^2s}{dt^2} = \frac{d}{dt} (\dot{\gamma}(t) \cdot \dot{\gamma}(t)) = 2\dot{\gamma}(t) \cdot \ddot{\gamma}(t),$$

i.e.,

$$\frac{d^2s}{dt^2} \frac{ds}{dt} = \dot{\gamma}(t) \cdot \ddot{\gamma}(t),$$

so we get

$$\kappa = \left\| \frac{\frac{d^2\gamma}{dt^2} \frac{ds}{dt} - \frac{d\gamma}{dt} \frac{d^2s}{dt^2}}{(ds/dt)^3} \right\| = \left\| \frac{\left(\frac{ds}{dt}\right)^2 \ddot{\gamma}(t) - \frac{d^2s}{dt^2} \frac{ds}{dt} \dot{\gamma}(t)}{(ds/dt)^4} \right\| = \left\| \frac{\left(\dot{\gamma}(t) \cdot \dot{\gamma}(t)\right) \ddot{\gamma}(t) - \left(\dot{\gamma}(t) \cdot \ddot{\gamma}(t)\right) \dot{\gamma}(t)}{\|\dot{\gamma}(t)\|^4} \right\|,$$

since we have

$$\dot{\gamma}(t) \times \left(\ddot{\gamma}(t) \times \dot{\gamma}(t) \right) = \left(\dot{\gamma}(t) \cdot \dot{\gamma}(t) \right) \ddot{\gamma}(t) - \left(\dot{\gamma}(t) \cdot \ddot{\gamma}(t) \right) \dot{\gamma}(t),$$

note that $\langle \dot{\gamma}(t), \ddot{\gamma}(t) \times \dot{\gamma}(t) \rangle = \frac{\pi}{2}$, so

$$\left\| \dot{\gamma}(t) \times \left(\ddot{\gamma}(t) \times \dot{\gamma}(t) \right) \right\| = \|\dot{\gamma}(t)\| \cdot \|\ddot{\gamma}(t) \times \dot{\gamma}(t)\|,$$

thus we finally see that the curvature κ is

$$\begin{aligned} \kappa &= \frac{\left\| \left(\dot{\gamma}(t) \cdot \dot{\gamma}(t) \right) \ddot{\gamma}(t) - \left(\dot{\gamma}(t) \cdot \ddot{\gamma}(t) \right) \dot{\gamma}(t) \right\|}{\|\dot{\gamma}(t)\|^4} = \frac{\left\| \dot{\gamma}(t) \times \left(\ddot{\gamma}(t) \times \dot{\gamma}(t) \right) \right\|}{\|\dot{\gamma}(t)\|^4} \\ &= \frac{\|\dot{\gamma}(t)\| \|\ddot{\gamma}(t) \times \dot{\gamma}(t)\|}{\|\dot{\gamma}(t)\|^4} = \frac{\|\ddot{\gamma}(t) \times \dot{\gamma}(t)\|}{\|\dot{\gamma}(t)\|^3}, \end{aligned}$$

which completes the proof. ■

Example 2.(Curvature of Circular Helix)

A *circular helix* is of the form

$$\gamma(\theta) := (a \cos \theta, a \sin \theta, b\theta), \quad \theta \in \mathbb{R},$$

where $a, b \in \mathbb{R}$ are constants. So we get

$$\dot{\gamma}(\theta) = (-a \sin \theta, a \cos \theta, b),$$

$$\ddot{\gamma}(\theta) = (-a \cos \theta, -a \sin \theta, 0),$$

thus we have

$$\|\dot{\gamma}(\theta)\| = \sqrt{a^2 + b^2},$$

$$\begin{aligned} \ddot{\gamma}(\theta) \times \dot{\gamma}(\theta) &= \left(\det \begin{pmatrix} -a \sin \theta & 0 \\ a \cos \theta & b \end{pmatrix}, -\det \begin{pmatrix} -a \cos \theta & 0 \\ -a \sin \theta & b \end{pmatrix}, \det \begin{pmatrix} -a \cos \theta & -a \sin \theta \\ -a \sin \theta & a \cos \theta \end{pmatrix} \right) \\ &= (-ab \sin \theta, ab \cos \theta, -a^2). \end{aligned}$$

so we get

$$\kappa(\theta) = \frac{\|(-ab \sin \theta, ab \cos \theta, -a^2)\|}{\|(-a \sin \theta, a \cos \theta, b)\|^3} = \frac{(a^2 b^2 + a^4)^{\frac{1}{2}}}{(a^2 + b^2)^{\frac{3}{2}}} = \frac{|a|}{a^2 + b^2}.$$

Remark: Pariticularly, in the case that $b = 0$, which means the circular helix is a plane circle of radius $|a|$, then $\kappa \equiv \frac{1}{|a|}$; in the case that $a = 0$, which means that it is actually a straight line, then $\kappa \equiv 0$.

1.2 Signed Curvature and Turning Angle of Plane Curves

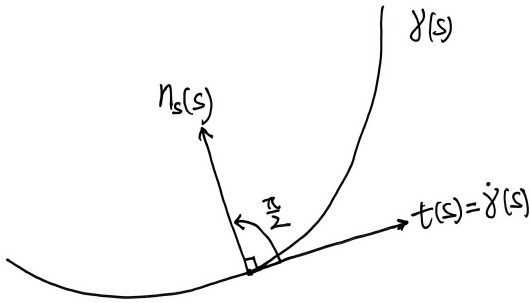
• Now we consider the case that γ is a plane curve. Assume $\gamma(s)$ is a unit-speed curve in $\mathbb{R}^2$. Now we use the dot symbol to denote $\frac{d}{ds}$.

Notation: Let $\gamma(s)$ be a plane curve with unit-speed, i.e., $\|\dot{\gamma}(s)\| \equiv 1$, we denote

$$\mathbf{t}(s) := \dot{\gamma}(s),$$

i.e., $\mathbf{t}(s)$ is the tangent vector of $\gamma(s)$, hence $\|\mathbf{t}(s)\| \equiv 1$.

Notation: We denote $\mathbf{n}_s$ to be the *signed unit normal vector* of γ , i.e., $\|\mathbf{n}_s(s)\| \equiv 1$ and $\mathbf{n}_s(s)$ is obtained by rotating $\mathbf{t}(s)$ anticlockwise by $\frac{\pi}{2}$.



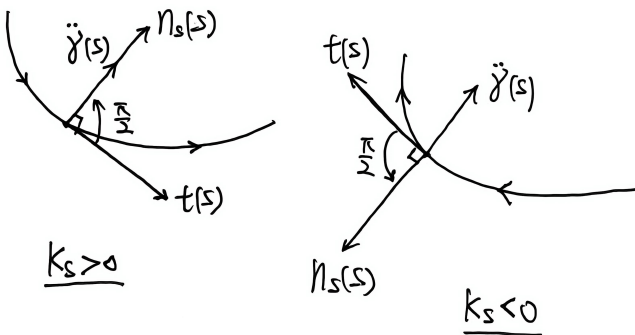
Note that $\|\mathbf{t}(s)\| \equiv 1$, thus $\dot{\mathbf{t}}(s) = \ddot{\gamma}(s) \perp \mathbf{t}(s) = \dot{\gamma}(s)$ for all s , which implies that $\dot{\mathbf{t}}(s) \parallel \mathbf{n}_s(s)$. So this leads to the following definition.

Definition 2. (Signed Curvature of a Plane Curve)

Let $\gamma(s)$ be a unit-speed plane curve, then there is a scalar function κ_s such that

$$\dot{\mathbf{t}}(s) = \ddot{\gamma}(s) \triangleq \kappa_s(s) \mathbf{n}_s(s),$$

κ_s is called *signed curvature* of γ .



Proposition 2. Let $\gamma(s)$ be a unit-speed plane curve, then we have

$$\kappa(s) = |\kappa_s(s)| \text{ for } \forall s,$$

i.e., the curvature of $\gamma(s)$ is the absolute value of its signed curvature.

Proof: Just note that

$$\kappa(s) = \|\ddot{\gamma}(s)\| = \|\kappa_s(s)\mathbf{n}_s(s)\| = |\kappa_s(s)|\|\mathbf{n}_s(s)\| = |\kappa_s(s)|,$$

since $\|\mathbf{n}_s(s)\| \equiv 1$. ■

If $\gamma(t)$ is a regular but not necessarily unit-speed curve, then we define $\mathbf{t}$ and $\mathbf{n}_s$ and κ_s to be those of its unit-speed reparametrization $\tilde{\gamma}(s)$, where s is the arc-length of γ .

Proposition 3. Let $\gamma(t)$ be a regular curve (not necessarily unit-speed), then we have

$$\frac{d\mathbf{t}}{dt} = \kappa_s \left\| \frac{d\gamma}{dt} \right\| \mathbf{n}_s.$$

Proof: We see that

$$\mathbf{t}(s) = \frac{d\tilde{\gamma}}{ds} = \frac{d\gamma}{dt} \frac{dt}{ds} = \frac{d\gamma/dt}{ds/dt} = \frac{d\gamma/dt}{\|d\gamma/dt\|},$$

so we get

$$\frac{d\mathbf{t}}{dt} = \frac{d\mathbf{t}}{ds} \frac{ds}{dt} = \kappa_s \mathbf{n}_s \frac{ds}{dt} = \kappa_s \left\| \frac{d\gamma}{dt} \right\| \mathbf{n}_s,$$

which is the equation. ■

2 Plane Curves

- Let $\gamma : (\alpha, \beta) \rightarrow \mathbb{R}^2$ be a unit-speed plane curve, then there is an angle $\varphi(s)$ satisfying

$$\dot{\gamma}(s) \triangleq (\cos \varphi(s), \sin \varphi(s))$$

for every point $\gamma(s)$. This angle $\varphi(s)$ is not unique, the following proposition tells us that there will be a unique one if some additional condition is required.

Proposition 4. Let $\gamma : (\alpha, \beta) \rightarrow \mathbb{R}^2$ be a unit-speed curve, let $s_0 \in (\alpha, \beta)$ and φ_0 be such that $\dot{\gamma}(s_0) = (\cos \varphi_0, \sin \varphi_0)$, then there is a unique function $\varphi \in C^\infty((\alpha, \beta); \mathbb{R})$ such that

$$\dot{\gamma}(s) = (\cos \varphi(s), \sin \varphi(s)) \text{ for } \forall s \in (\alpha, \beta)$$

and $\varphi(s_0) = \varphi_0$.

Proof: Denote $\dot{\varphi}(s) \triangleq (f(s), g(s))$, thus $f(s_0) = \cos \varphi_0$, $g(s_0) = \sin \varphi_0$ and $f(s)^2 + g(s)^2 \equiv 1$ for $\forall s \in (\alpha, \beta)$ since $\|\dot{\gamma}(s)\| \equiv 1$. Define the function $\varphi(s)$ as

$$\varphi(s) := \varphi_0 + \int_{s_0}^s (f(t)\dot{g}(t) - \dot{f}(t)g(t))dt,$$

which is clear that $\varphi(s_0) = \varphi_0$ and $\varphi(s)$ is smooth;

Denote

$$F(s) := f(s) \cos \varphi(s) + g(s) \sin \varphi(s);$$

$$G(s) := f(s) \sin \varphi(s) - g(s) \cos \varphi(s).$$

So we get

$$\dot{F}(s) = \left(\dot{f}(s) + g(s)\dot{\varphi}(s) \right) \cos \varphi(s) + \left(\dot{g}(s) - f(s)\dot{\varphi}(s) \right) \sin \varphi(s),$$

note that

$$\dot{\varphi}(s) = f(s)\dot{g}(s) - \dot{f}(s)g(s),$$

so $\dot{f} + g\dot{\varphi} = \dot{f} + g(f\dot{g} - \dot{f}g) = \dot{f}(1 - g^2) + fg\dot{g} = \dot{f}f^2 + fg\dot{g} = f(f\dot{f} + g\dot{g}) = 0$ since we see that $\frac{d}{ds}(f(s)^2 + g(s)^2) \equiv 0$. So this shows that

$$\dot{F}(s) \equiv 0 \quad \text{and hence } F \equiv \text{constant}.$$

Another similar argument shows that

$$\dot{G}(s) \equiv 0 \quad \text{and hence } G \equiv \text{constant}.$$

Note that

$$F(s_0) = f(s_0) \cos \varphi(s_0) + g(s_0) \sin \varphi(s_0) = \cos^2 \varphi_0 + \sin^2 \varphi_0 = 1,$$

$$G(s_0) = f(s_0) \sin \varphi(s_0) - g(s_0) \cos \varphi(s_0) = \cos \varphi_0 \sin \varphi_0 - \sin \varphi_0 \cos \varphi_0 = 0.$$

So we see that $F(s) \equiv 1$ and $G(s) \equiv 0$, which gives us the equations

$$\begin{cases} f(s) \cos \varphi(s) + g(s) \sin \varphi(s) \equiv 1, \\ f(s) \sin \varphi(s) - g(s) \cos \varphi(s) \equiv 0. \end{cases}$$

which implies that $f(s) = \cos \varphi(s)$, $g(s) = \sin \varphi(s)$;

If $\psi(s) \in C^\infty((\alpha, \beta); \mathbb{R})$ is another smooth angle function such that $\dot{\gamma}(s) = (\cos \psi(s), \sin \psi(s))$ for $\forall s \in (\alpha, \beta)$ and $\psi(s_0) = \varphi_0$, then $\exists n(s) \in \mathbb{Z}$ such that

$$\psi(s) = \varphi(s) + 2n(s)\pi \quad \text{for } \forall s \in (\alpha, \beta)$$

i.e.,

$$n(s) = \frac{\psi(s) - \varphi(s)}{2\pi} \in C^\infty((\alpha, \beta); \mathbb{Z}),$$

note that

$$n : (\alpha, \beta) \longrightarrow \mathbb{Z}$$

is a continuous map and the interval (α, β) is connected so $n((\alpha, \beta)) \subseteq \mathbb{Z}$ must be connected, which means that $n(s)$ is a constant integer and independent of s . Note that $n(s_0) = \frac{\psi(s_0) - \varphi(s_0)}{2\pi} = 0$, so $n(s) \equiv 0$. ■

The angle $\varphi(s)$ in the proposition above gives the following definition.

Definition 3.(Turning Angle)

The angle function $\varphi \in C^\infty((\alpha, \beta); \mathbb{R})$ in the Proposition 4 of the plane curve $\gamma : (\alpha, \beta) \longrightarrow \mathbb{R}^2$ is called the **turning angle** of γ determined by the condition $\varphi(s_0) = \varphi_0$.

The next proposition tells us that the signed curvature of a plane curve is actually the rate at which the tangent vector of the curve rotates.

Proposition 5. Let $\gamma(s)$ be a unit-speed plane curve and let $\varphi(s)$ be a turning angle for γ , then we have

$$\kappa_s(s) = \frac{d\varphi}{ds}.$$

Proof: By our notation, we have

$$\mathbf{t}(s) = \dot{\gamma}(s) = \begin{pmatrix} \cos \varphi(s) \\ \sin \varphi(s) \end{pmatrix},$$

hence

$$\dot{\mathbf{t}}(s) = \dot{\varphi}(s) \begin{pmatrix} -\sin \varphi(s) \\ \cos \varphi(s) \end{pmatrix}.$$

Note that $\mathbf{n}_s(s) = \begin{pmatrix} -\sin \varphi(s) \\ \cos \varphi(s) \end{pmatrix}$, so by the definition $\ddot{\gamma}(s) = \kappa_s \mathbf{n}_s(s)$, we get $\kappa_s(s) = \dot{\varphi}(s)$. ■

Remark: If γ is a unit-speed closed plane curve, then the **total signed curveture** of γ is given by

$$\int_0^l \kappa_s(s) ds, \text{ where } l \text{ is the length of } \gamma.$$

Proposition 6. (Total Signed Curvature of a Closed Plane Curve)

If γ is a unit-speed closed plane curve, then $\exists k \in \mathbb{Z}^+$ such that

$$\int_0^l \kappa_s(s) ds = 2k\pi.$$

Proof: By Proposition 5, we know that $\kappa_s(s) = \frac{d\varphi}{ds}$, hence

$$\int_0^l \kappa_s(s) ds = \int_0^l \frac{d\varphi}{ds} ds = \varphi(l) - \varphi(0).$$

Note that $\gamma(s)$ is l -periodic, i.e.,

$$\gamma(s+l) \equiv \gamma(s) \text{ for } \forall s,$$

so differentiating both sides w.r.t. s , we have

$$\dot{\gamma}(s+l) \equiv \dot{\gamma}(s) \text{ for } \forall s,$$

which gives us that $\dot{\gamma}(l) = \dot{\gamma}(0)$, which means

$$\begin{pmatrix} \cos \varphi(l) \\ \sin \varphi(l) \end{pmatrix} = \begin{pmatrix} \cos \varphi(0) \\ \sin \varphi(0) \end{pmatrix},$$

which means that $\exists k \in \mathbb{Z}^+$ such that $\varphi(l) - \varphi(0) = 2k\pi$. ■

Notation: We denote the anticlockwise rotation by an angle θ as ρ_θ , i.e.,

$$\begin{aligned} \rho &: \mathbb{R} \longrightarrow \text{GL}_2(\mathbb{R}) \\ \theta &\mapsto \rho_\theta := \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}. \end{aligned}$$

Notation: We denote the translation by a constant vector $\mathbf{a}$ by $T_{\mathbf{a}}$, i.e.,

$$T_{\mathbf{a}} : \mathbf{v} \mapsto \mathbf{v} + \mathbf{a} \text{ for } \forall \mathbf{v} \in \mathbb{R}^2.$$

By these two notations above, an **isometry** in $\mathbb{R}^2$ means a linear map

$$M : \mathbb{R}^2 \longrightarrow \mathbb{R}^2$$

such that $M \triangleq \rho_\theta \circ T_{\mathbf{a}}$ for some angle $\theta \in \mathbb{R}$ and some constant vector $\mathbf{a} \in \mathbb{R}^2$.

The following theorem tells us that unit-speed plane curves are determined by their signed curvature at every point up to an isometry of $\mathbb{R}^2$.

Theorem 1. (Plane Curves Are Determined by Signed Curvature up to Isometry)

For $\forall k(s) \in C^\infty((\alpha, \beta); \mathbb{R})$, there is a unit-speed plane curve $\gamma : (\alpha, \beta) \longrightarrow \mathbb{R}^2$ whose signed curvature is $k(s)$, if $\tilde{\gamma}(s)$ is another unit-speed plane curve whose signed curvature is $k(s)$ as well, then there is an isometry M of $\mathbb{R}^2$ (i.e., $M \in \text{ISO}(2)$) such that

$$\tilde{\gamma}(s) = M(\gamma(s)) \text{ for } \forall s \in (\alpha, \beta)$$

Proof: For any fixed $s_0 \in (\alpha, \beta)$, we define

$$\varphi(s) := \int_{s_0}^s k(t) dt \text{ for } \forall s \in (\alpha, \beta)$$

and define

$$\gamma(s) := \left(\int_{s_0}^s \cos \varphi(t) dt, \int_{s_0}^s \sin \varphi(t) dt \right).$$

Thus we see that

$$\dot{\gamma}(s) = (\cos \varphi(s), \sin \varphi(s)).$$

Hence $\|\dot{\gamma}(s)\| \equiv 1$ and by Proposition 5, we know that its signed curvature is

$$\kappa_s(s) = \frac{d\varphi}{ds} = k(s);$$

If $\tilde{\gamma}(s)$ is another such curve, then we denote the smooth turning angle for $\tilde{\gamma}(s)$ as $\tilde{\varphi}(s)$. i.e.,

$$\dot{\tilde{\gamma}}(s) = (\cos \tilde{\varphi}(s), \sin \tilde{\varphi}(s)),$$

so we have

$$\tilde{\gamma}(s) = \left(\int_{s_0}^s \cos \tilde{\varphi}(t) dt, \int_{s_0}^s \sin \tilde{\varphi}(t) dt \right) + \tilde{\gamma}(s_0).$$

Since

$$\frac{d\tilde{\varphi}}{ds} = k(s),$$

so

$$\tilde{\varphi}(s) = \int_{s_0}^s k(u) du + \tilde{\varphi}(s_0).$$

So we have the computation as below

$$\begin{aligned}
\tilde{\gamma}(s) &= T_{\tilde{\gamma}(s_0)} \left(\int_{s_0}^s \cos \tilde{\varphi}(t) dt, \int_{s_0}^s \sin \tilde{\varphi}(t) dt \right) \\
&= T_{\tilde{\gamma}(s_0)} \left(\int_{s_0}^s \cos \tilde{\varphi}(t) dt, \int_{s_0}^s \sin \tilde{\varphi}(t) dt \right) \\
&= T_{\tilde{\gamma}(s_0)} \left(\int_{s_0}^s \cos \left(\int_{s_0}^t k(u) du + \tilde{\varphi}(s_0) \right) dt, \int_{s_0}^s \sin \left(\int_{s_0}^t k(u) du + \tilde{\varphi}(s_0) \right) dt \right) \\
&= T_{\tilde{\gamma}(s_0)} \left(\int_{s_0}^s \cos \left(\int_{s_0}^t \frac{d\varphi}{du} du + \tilde{\varphi}(s_0) \right) dt, \int_{s_0}^s \sin \left(\int_{s_0}^t \frac{d\varphi}{du} du + \tilde{\varphi}(s_0) \right) dt \right) \quad (\text{because } k(u) = \frac{d\varphi}{du}) \\
&= T_{\tilde{\gamma}(s_0)} \left(\int_{s_0}^s \cos (\varphi(t) + \tilde{\varphi}(s_0)) dt, \int_{s_0}^s \sin (\varphi(t) + \tilde{\varphi}(s_0)) dt \right) \quad (\text{note that } \varphi(s_0) = 0) \\
&= T_{\tilde{\gamma}(s_0)} \left(\cos \tilde{\varphi}(s_0) \int_{s_0}^s \cos \varphi(t) dt - \sin \tilde{\varphi}(s_0) \int_{s_0}^s \sin \varphi(t) dt, \right. \\
&\quad \left. \sin \tilde{\varphi}(s_0) \int_{s_0}^s \cos \varphi(t) dt + \cos \tilde{\varphi}(s_0) \int_{s_0}^s \sin \varphi(t) dt \right) \\
&= T_{\tilde{\gamma}(s_0)} \left(\rho_{\tilde{\varphi}(s_0)} \left(\int_{s_0}^s \cos \varphi(t) dt, \int_{s_0}^s \sin \varphi(t) dt \right) \right) \\
&= (T_{\tilde{\gamma}(s_0)} \circ \rho_{\tilde{\varphi}(s_0)}) \left(\int_{s_0}^s \cos \varphi(t) dt, \int_{s_0}^s \sin \varphi(t) dt \right) = (T_{\tilde{\gamma}(s_0)} \circ \rho_{\tilde{\varphi}(s_0)}) (\gamma(s)).
\end{aligned}$$

So the isometry $M \in \text{ISO}(2)$ is given by $M = T_{\tilde{\gamma}(s_0)} \circ \rho_{\tilde{\varphi}(s_0)}$. ■

3 Appendix: Results from Vector Algebra

Proposition 7. Let $\mathbf{a}, \mathbf{b}, \mathbf{c}$ be three vectors in $\mathbb{R}^3$, then we have the following equality

$$\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = (\mathbf{a} \cdot \mathbf{c})\mathbf{b} - (\mathbf{a} \cdot \mathbf{b})\mathbf{c}.$$

Proof: Let $[O; e_1, e_2, e_3]$ be a right-handed Cartesian coordinate system, suppose that the coordinates for $\mathbf{a}, \mathbf{b}, \mathbf{c}$ are

$$\mathbf{a} \triangleq (a_1, a_2, a_3), \quad \mathbf{b} \triangleq (b_1, b_2, b_3), \quad \mathbf{c} \triangleq (c_1, c_2, c_3)$$

respectively under this system. Assume that we express $\mathbf{b} \times \mathbf{c}$ and $\mathbf{a} \times (\mathbf{b} \times \mathbf{c})$ as (d_1, d_2, d_3) , (h_1, h_2, h_3) respectively. Then we know that

$$\begin{aligned}
h_1 &= \det \begin{pmatrix} a_2 & a_3 \\ d_2 & d_3 \end{pmatrix} = a_2 d_3 - a_3 d_2 \\
&= a_2 \det \begin{pmatrix} b_1 & b_2 \\ c_1 & c_2 \end{pmatrix} - a_3 \det \begin{pmatrix} b_3 & b_1 \\ c_3 & c_1 \end{pmatrix} \\
&= a_2(b_1 c_2 - b_2 c_1) - a_3(b_3 c_1 - b_1 c_3) \\
&= b_1(a_2 c_2 + a_3 c_3) - c_1(a_2 b_2 + a_3 b_3) \\
&= b_1(\mathbf{a} \cdot \mathbf{c} - a_1 c_1) - c_1(\mathbf{a} \cdot \mathbf{b} - a_1 b_1) \\
&= (\mathbf{a} \cdot \mathbf{b})b_1 - (\mathbf{a} \cdot \mathbf{c})c_1.
\end{aligned}$$

Similarly, we have

$$h_2 = (\mathbf{a} \cdot \mathbf{b})b_2 - (\mathbf{a} \cdot \mathbf{b})c_2, \quad h_3 = (\mathbf{a} \cdot \mathbf{b})b_3 - (\mathbf{a} \cdot \mathbf{b})c_3.$$

Therefore we get

$$\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = (\mathbf{a} \cdot \mathbf{c})\mathbf{b} - (\mathbf{a} \cdot \mathbf{b})\mathbf{c},$$

which is the desired equality. ■